ClipSAT : Differential Equations First/Second Order, Laplace, Systems & S

Unit 8.01: Power Series Solutions & Method of Frobenius

Name: _____

Q1. For x^{2} - xy''+y'-y=0 , the point x=0 is:

[ ] A: ordinary point
[ ] B: a point where no solution exists
[ ] C: singular point
[ ] D: a point where every solution vanishe

Q2. Near x_0=0 , the power series solutions of x + 4y''+xy'+y=0 are guaranteed to converge at least for |x-x_0|<R with R=

[ ] A: 5
[ ] B: 4
[ ] C: infty
[ ] D: 1

Q3. For x^{2} - 9y''+y'-y=0 , the point x=-1 is:

[ ] A: singular point
[ ] B: ordinary point
[ ] C: a point where no solution exists
[ ] D: a point where every solution vanishe

Q4. Near x_0=0 , the power series solutions of x^{2} - 4 x + 3y''+xy'+y=0 are guaranteed to converge at least for |x-x_0|<R with R=

[ ] A: 3
[ ] B: infty
[ ] C: 1
[ ] D: 2

Q5. For x^{2} + xy''+y'-y=0 , the point x=-1 is:

[ ] A: singular point
[ ] B: ordinary point
[ ] C: a point where no solution exists
[ ] D: a point where every solution vanishe

Q6. Near x_0=0 , the power series solutions of x^{2} + 4y''+xy'+y=0 are guaranteed to converge at least for |x-x_0|<R with R=

[ ] A: 2
[ ] B: infty
[ ] C: 1
[ ] D: 4

Q7. For x^{2} - 4 xy''+y'-y=0 , the point x=0 is:

[ ] A: ordinary point
[ ] B: a point where no solution exists
[ ] C: a point where every solution vanishe
[ ] D: singular point

Q8. Near x_0=0 , the power series solutions of x^{2} + x - 6y''+xy'+y=0 are guaranteed to converge at least for |x-x_0|<R with R=

[ ] A: 2
[ ] B: 3
[ ] C: infty
[ ] D: 1